Ezairo® 7100 Errata

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Problem Identifier	Problem Description	Impact	Workaround
Coupling between LSAD and the input stage.	Common-mode coupling at the LSAD sampling frequency has been observed. This coupling has been confirmed as entering the system after the preamplifiers.	Tones can be observed in the input audio at multiples of the LSAD sampling rate.	The LSADs can be used at a burst rate that is either non-periodic or is at a frequency outside of the audio frequency range of interest. A sample implementation of this workaround is provided as part of the Battery Monitor sample application.
CFX writes to ARM Cortex-M3 processor's program memory can be corrupted when the ARM Cortex-M3 processor clock is divided.	The ARM Cortex-M3 processor's program memory can be corrupted when: • The ARM Cortex-M3 processor clock is divided (`CLK_CFG_0_CM3CLK_PRESCALE` bit field from the `D_CLK_CFG_0` register has a non-zero value) • The ARM Cortex-M3 processor is writing to or reading from program memory in consecutive clock cycles • The CFX is reading or writing the ARM Cortex-M3 processor's program memory If all three conditions hold, the address and size (byte, half-word or word length) of the ARM Cortex-M3 processor's program memory access is undefined.	If this problem occurs, the memory read or written by the ARM Cortex-M3 processor is undefined and likely to be incorrect.	Workaround options include: • Not dividing the ARM Cortex-M3 processor clock • Restricting CFX processor accesses to the ARM Cortex-M3 processor's data memory while the ARM Cortex-M3 processor is executing. • The use of interrupts to create a handshaking scheme where the ARM Cortex-M3 processor indicates when the CFX can safely access its program memory.
Undesired GPIO interrupts when GPIO interrupt source 0 is not enabled first.	When a GPIO interrupt source other than source 0 is enabled first, an unexpected interrupt can occur. This problem applies equally to the CFX GPIO interrupts and ARM Cortex-M3 core's GPIO interrupts.	An undesired GPIO interrupt can occur in the first four cycles after an interrupt source has been enabled if both of the following are true: • The GPIO interrupt source is not interrupt source 0. • No GPIO interrupt sources are currently enabled.	Workaround options include: • Use GPIO (C_GPIO) interrupt source 0. • Ignore GPIO interrupts during the first four cycles after enabling a GPIO interrupt source other than 0, if no other sources were previously enabled. • Ensure that a GPIO interrupt source is always enabled.

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The AES decryption routines use a portion of CD0 memory.	The AES routines unpack their SBOX and inverse SBOX tables to an area of CD0 memory mapped into the X memory space between 0xA400 and 0xA5FF. This has no effect on bootloading applications, as the tools ensure that all P memory sections (including all encrypted P memory sections) are loaded prior to any X memory sections. However, if an application uses the AES decryption routines, this area cannot be used by other components of the application.	If you are using the AES decryption routines, consider the area between X:0xA400 and X:0xA5FF as firmware reserved memory.	Reserve the X:0xA400 to X:0xA5FF portion of CD0 memory in any application that uses the AES decryption routines after booting.
The EEPROM library linked into the Program ROM cannot write multiple EEPROM pages above 0xFFFF.	If the EEPROM is 24-bit addressable, block writes using the Program ROM's EEPROM library that span two pages, where the second page starts at 0x10000 or higher, fail to write to the correct second page.	The EEPROM library has been updated to resolve this issue; however this implementation consumes more code space because it cannot reuse the Program ROM EEPROM library implementation.	Use EEPROM library 3.2.0 which does not use the Program ROM's EEPROM library instance for block write functions. If a user application is not writing more than one page above 0x10000, it can use the strict ROM vector version of the library (*sk4_eeprom_strict_rom_cid*.a*) to avoid using additional code space to support EEPROM block writes.
The `PC` and `LR` are loaded incorrectly when single-stepping through a `CALL` instruction that targets an address containing a `GOTO` instruction, while one or more interrupts are pending and interrupts are disabled.	When an enabled interrupt is pending and interrupts are disabled (the `IE` bit in the status register is 0), then single-stepping over a `CALL` instruction whose first target instruction is a `GOTO` behaves incorrectly. In this scenario, the `PC` register is set to the target address of the `GOTO` instruction (and not to the address of the `GOTO` instruction itself), and the `LR` register is incorrectly updated to the address of the `GOTO` instruction +1.	This behavior is exhibited only in debug mode, and only when single-stepping the `CALL` instruction that points to a `GOTO` instruction. When this combination occurs, this results in the incorrect execution of a program. When the core is running continuously, no issues are observed.	You can simulate a single-step over a `CALL` instruction that calls a `GOTO` instruction by placing a breakpoint on the `GOTO` instruction and then running until the DSP hits the breakpoint.

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